SENTHIL PUBLIC SCHOOL • CBSE GRADED ANSWER SCRIPT

Exam ID: **EXAM-MAT-DAI-20261018-664F52** | Student: **Tharun** | Subject: **Mathematics** | Score: **17/17 (100.0%)**

Q.1. Evaluate using suitable identity: (103)^2 = (100 + 3)^2. [Awarded: 1/1M]

Tharun's Working:
(a) 10,609
Working: Using algebraic identity (a + b)^2 = a^2 + 2ab + b^2
(100 + 3)^2 = 100^2 + 2(100)(3) + 3^2 = 10000 + 600 + 9 = 10,609.
Correct option is (a).

Teacher Red-Ink Feedback: Flawless step execution and correct option.

Q.2. The product of monomials (-4p) and (7pq) is: [Awarded: 1/1M]

Tharun's Working:
(b) -28p^2q
Working: Product of monomials = (-4p) * (7pq) = (-4 * 7) * (p * p * q) = -28p^2q.
Correct option is (b).

Teacher Red-Ink Feedback: Flawless step execution and correct option.

Q.3. Multiply the binomials: (2x + 5) and (4x - 3). Show all steps. [Awarded: 2/2M]

Tharun's Working:
Multiply binomials: (2x + 5) and (4x - 3)
Step 1: Apply distributive property
= 2x * (4x - 3) + 5 * (4x - 3)
= (2x * 4x) - (2x * 3) + (5 * 4x) - (5 * 3)
= 8x^2 - 6x + 20x - 15
Step 2: Combine like terms (-6x + 20x = 14x)
= 8x^2 + 14x - 15.
Final Answer Box: [8x^2 + 14x - 15]

Teacher Red-Ink Feedback: Clear step separation and accurate algebraic simplification.

Q.4. Solve the linear equation: 5x + 9 = 5 + 3x. [Awarded: 2/2M]

Tharun's Working:
Solve linear equation: 5x + 9 = 5 + 3x
Step 1: Transposing variables to LHS and constants to RHS
5x - 3x = 5 - 9
2x = -4
Step 2: Dividing by 2
x = -4 / 2
x = -2.
Final Answer Box: [x = -2]

Teacher Red-Ink Feedback: Clear step separation and accurate algebraic simplification.

Q.5. Find the area of a trapezium whose parallel sides are 12 cm and 20 cm, and the distance between them is 8 cm. [Awarded: 3/3M]

Tharun's Working:
Area of Trapezium
Given: Parallel sides a = 12 cm, b = 20 cm, height h = 8 cm.
Formula: Area = 1/2 * (a + b) * h
Step 1: Substitute values
Area = 1/2 * (12 + 20) * 8
Area = 1/2 * 32 * 8
Area = 16 * 8 = 128 cm^2.
Final Answer Box: [128 cm²]

Teacher Red-Ink Feedback: Exemplary presentation. Full marks for intermediate steps.

Q.6. Simplify using algebraic identity: (2a - 7)^2. [Awarded: 3/3M]

Tharun's Working:
Simplify using algebraic identity: (2a - 7)^2
Identity: (x - y)^2 = x^2 - 2xy + y^2
Here x = 2a, y = 7.
Step 1: (2a)^2 - 2(2a)(7) + (7)^2
Step 2: 4a^2 - 28a + 49.

Final Answer Box: [4a^2 - 28a + 49]

Teacher Red-Ink Feedback: Exemplary presentation. Full marks for intermediate steps.

Q.7. A rectangular sheet of paper 44 cm x 20 cm is rolled along its length to form a cylinder. Find the volume of the cylinder. (Take pi = 22/7) [Awarded: 5/5M]

Tharun's Working:
Cylinder rolled from rectangular sheet 44 cm x 20 cm
Rolled along length => Length becomes circumference of base, width becomes height.
Circumference C = 2 * pi * r = 44 cm
2 * (22/7) * r = 44
(44/7) * r = 44 => r = 7 cm.
Height h = 20 cm.
Formula for Volume of Cylinder = pi * r^2 * h
Volume = (22/7) * (7)^2 * 20
Volume = (22/7) * 49 * 20
Volume = 22 * 7 * 20 = 154 * 20 = 3080 cm^3.
Final Answer Box: [Volume = 3080 cm^3]

Teacher Red-Ink Feedback: Outstanding multi-step derivation. Fully adheres to CBSE 100/100 marking criteria.